INCH-POUND
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PERFORMANCE SPECIFICATION

POWER UNIT, HYDRAULIC

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hydraulic power unit which is a self-contained electrically driven assembly capable of supplying not less than 2 gallons per minute (gal/min) of hydraulic fluid at 500 pounds per square inch gage (psig) and not less than 1 gal/min at 2000 psig. The hydraulic power unit will be referred to by the term, “power unit” (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4320

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-H-46170	- Hydraulic Fluid, Rust Inhabitant, Fire Resistant, Synthetic Hydrocarbon Base.
MIL-H-83282	- Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurements.

DEPARTMENT OF DEFENSE

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12293172	- Ramp Lock Installation.
12298252	- Power Supply, Hydro-electrical.
12359539	- Filter, EMI.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Part 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be in accordance with the manufacturer's materials specifications for hydraulic power unit and shall be capable of meeting all of the performance requirements specified herein. Materials shall be free of defects which adversely affect performance or serviceability of the finished product. Asbestos, cadmium, magnesium and radioactive material shall not be used in this item. Radioactive materials are defined by CFR, Title 10, Part 40, and other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item (see 4.6.1).

3.2.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion by using the manufacturer's standard process (see 4.6.1 and 4.6.2).

3.3 Design and construction. The power unit shall be configured in accordance with the tolerances of the size limits (FORM), interfaces and connections (FIT), shown on Army reference Drawing 12298252 and function in accordance with all of the performance requirements specified herein.

3.3.1 Interchangeability. All subassemblies and components having the same part number shall be functionally and dimensionally interchangeable (see 4.6.2).

3.3.2 Metals. All metals shall be compatible with the temperature and with the functional, service and storage conditions to which the components shall be exposed. The metals shall be suitably protected (see 4.6.1).

3.3.3 Hydraulic fluid. The power unit shall be compatible with fluids conforming to MIL-H-6083, MIL-H-5606, MIL-H-46170 and MIL-H-83282 (see 4.6.2).

3.3.4 Insulation. Electrical components shall be insulated from case structure. The power unit shall not be grounded internally (see 4.6.2).

3.3.5 High pressure filter. The pump discharge line shall include a 10 micron or finer filter (see 4.6.2).

3.3.6 Electric power. The pump and solenoid valve shall operate on 18 to 30 volt direct current (Vdc) with a current draw of no more than 140 amperes (A) to develop a pressure of 2000 psig. Electrical connections shall be as shown on Drawing 12298252 (see 4.6.1 and 4.6.2).

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3.3.7 Weight. The power unit shall weigh no more than 55 pounds (lb) dry weight (see 4.6.2).

3.3.8 Protective coating. All exposed components shall be treated to resist corrosion (see 4.6.1 and 4.6.2).

3.4 Performance. During the performance of the foregoing as well as environmental inspections, leaks and drips shall not be permissible. Weeps or seeps shall be allowable (see 6.3).

3.4.1 Fluid output. The power unit shall deliver no less than 2 gal/min at 500 psig and no less than 1 gal/min at 2000 psig (see 4.6.3).

3.4.1.1 Relief valve. The relief valve shall start to open (crack) at 2000 ±100 psig. Full volume flow shall be provided at 2250 ±250 psig (see 4.6.3.1).

3.4.1.2 Leakage. The power unit under 500 psig pressure shall leak no more than 0.2 cubic inches (in^3) of fluid in 10 seconds before and after meeting requirement 3.4.5. Under normal operation weeps and seeps shall be permissible and leaks or drips shall not be permissible (see 4.6.3.2 and 6.3).

3.4.2 Slope operation. The power unit shall be capable of operating on longitudinal and lateral slopes of ±66 percent (%). When mounted in longitudinal or lateral directions, no fluid shall leak from the power unit (see 4.6.4).

3.4.3 Solenoid valve operation. When actuated by an 18 to 30 Vdc signal, the solenoid valve shall reverse the flow between the outlet and discharge ports. The valve shall shift back to the original position when electrical power is removed (see 4.6.5).

3.4.4 Electromagnetic interference (EMI). The broadband emissions of the power unit in the frequency range of 0.15 to 1000 megahertz (MHz) shall not be generated and radiated in excess of the values shown on figure 1, except that the limit in the frequency range of 0.15 to 400 MHz shall be relaxed by 20 decibels (dB). The limits may be exceeded by no more than 35 dB in the frequency range of 2 MHz to 30 MHz. In order to conform to this requirement, an EMI filter, part number 12359539, may be used (see 4.6.6). Conducted switching spikes of less than 50 microseconds in duration shall not exceed the following, as applicable:

a. AC leads: ±50 percent of nominal root mean square (rms) voltage.
b. DC leads: +50 percent, -150 percent of nominal line voltage.

Spike duration is the time interval between the 50% amplitude point on the transient leading edge and the 50% amplitude point on the transient trailing edge; high frequency ringing superimposed on the pulse leading or trailing edges should be ignored.

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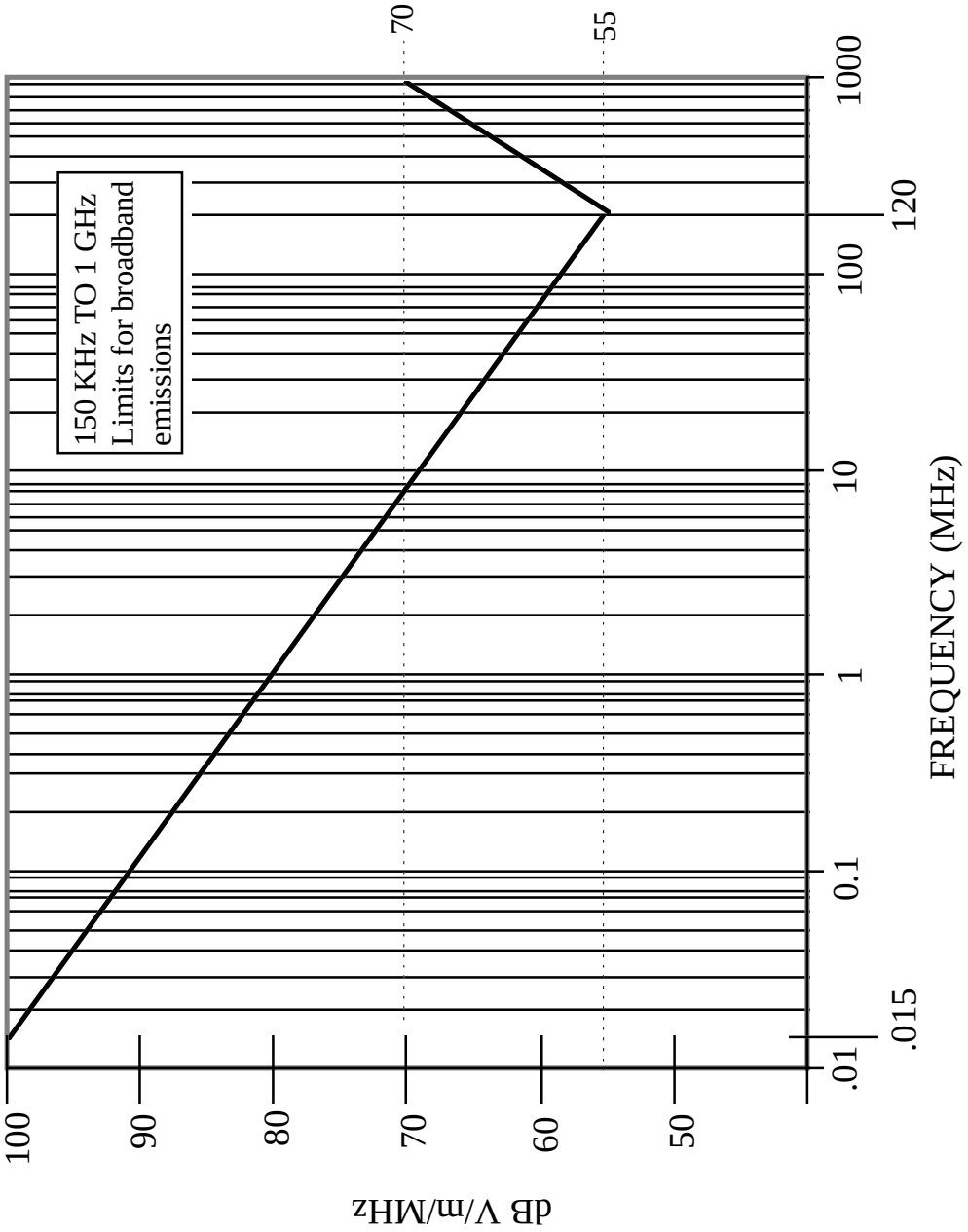


FIGURE 1 Limits for broadband emissions.

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3.4.5 Endurance. The hydraulic power unit shall operate the ramp or simulated ramp, satisfactorily for 1500 cycles from the fully extended to full retracted positions of the ramp. Total external leakage during the endurance test shall be no greater than 15% of the reservoir fluid volume prior to the test. Fluid shall be added and recorded as necessary during endurance. Following this test the number of cycles completed in 5 minutes shall not be less than 80% of those at the beginning of the test (see 4.6.7).

3.4.6 Environmental conditions.

3.4.6.1 High temperature. After storage of 160 degrees Fahrenheit (°F) for a period of 48 hours, the power unit shall be capable of delivering no less than 0.4 gal/min at 2000 psig at an operating temperature of 160°F. After operation at 160°F the power unit shall not exhibit structural damage of deformation and shall meet the requirements of 3.4.1 through 3.4.5. The power unit shall also meet the performance requirements of 3.4.1 when operated intermittently (see 6.3) at 125°F for a period of 30 minutes (see 4.6.8.1).

3.4.6.2 Low temperature. After storage at -70°F, the power unit shall operate at 65°F and shall meet the performance requirements of 3.4.1 through 3.4.3 (see 4.6.8.2).

3.4.6.3 Humidity. After exposure to 10 cycles of 95% relative humidity at a temperature of 86°F, the power unit shall meet the performance requirements of 3.4.1 through 3.4.3 at the conclusion of the fifth and tenth cycle (see 4.6.8.3).

3.4.6.4 Steam and waterjet cleaning. The power unit shall meet the performance requirements of 3.4.1 through 3.4.3 after being subjected to the steam and waterjet cleaning specified in 4.6.8.4.

3.4.6.5 Shock. The power unit shall meet the requirements of 3.4.1 after application of the following shock loading (see 4.6.8.5):

a. Power unit mounting - As shown on Drawing 12293172.
b. Area of shock application - At opposite face of shock mount from power unit.

3.4.6.6 Vibration. The power unit shall meet the requirements of 3.4.1 after the following vibration (see 4.6.8.6):

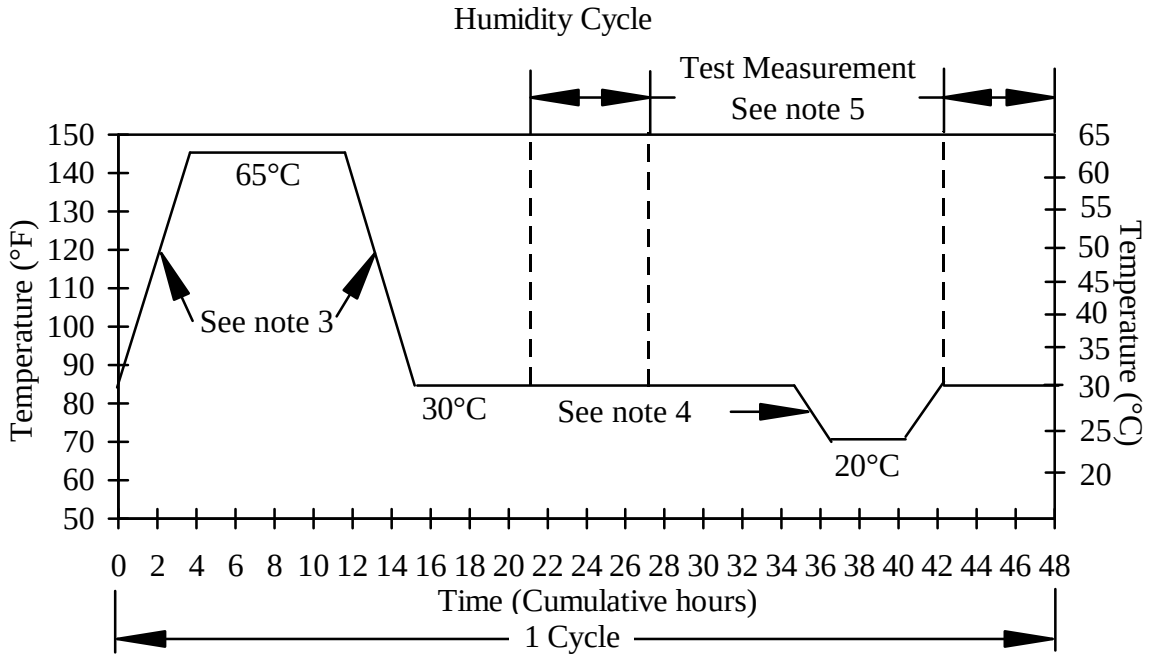
a. Power unit mounting - As shown on Drawing 12293172.
b. Area of vibration application - At opposite face of shock mount from power unit.
c. Level as shown in figure 2.

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3.4.6.7 Fungus resistance. The unit shall meet requirements after being subjected to the fungus test for 90 days as described in MIL-F-13927, class 2, method B (see 4.6.8.7).

3.4.6.8 Corrosion resistance. The unit shall meet requirements after exposure to the salt test in accordance with MIL-STD-202, test condition A of method 101D (see 4.6.8.8).

3.4.6.9 Sand and dust. The unit shall meet requirements after exposure to the sand and dust test in accordance with MIL-STD-202, method 110A (see 4.6.8.9).



NOTES:

1. Tolerance during temperature change shall be not greater than 5°F (3°C).
2. Relative humidity shall be maintained at 94 ±4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86 and 149°F (30 and 65°C) shall be not less than 14.4°F (8°C) per hour.
4. The temperature increase in this portion of the curve shall be not less than 18°F (10°C).
5. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 2. Humidity cycle.

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3.5 Finish. All exposed ferrous and aluminum surfaces shall be cleaned, treated, primed and finished (see 4.6.2). The film thickness of the primer shall be not less than 1 mil. The dry film thickness of the coating shall be not less than 1 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

3.6 Marking. The power unit shall be marked in accordance with the manufacture practice (see 4.6.2).

3.7 Workmanship. Workmanship shall be such as to provide a power unit free from defects such as cracked, dented, loose, broken or corroded parts and compartments, burrs or sharp edges and scratched finish (see 4.6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (CI) (see 4.5).

4.2 First article inspection. First article inspections shall be performed on preproduction and initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply hydraulic power units that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.2.1 Preproduction inspection (PPI). Unless otherwise specified (see 6.2), the preproduction inspection sample shall consist of two power units. The sample units shall be subjected to the PPIs specified in table I.

4.2.2 Initial production inspection (IPI). Unless otherwise specified (see 6.2), the initial production sample shall consist of three power units randomly selected from the first production lot manufactured. The sample units shall be subjected to the IPIs specified in table I.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 $\pm$18°F.
b. Barometric pressure: 28.5 (+2, - 3) inches Hg of mercury.
c. Relative humidity: 35 $\pm$5 percent.
d. Voltage: 27.5 $\pm$0.5 Vdc.

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4.3.1 Test conditions. Test specimens shall be stabilized at room temperature for one hour before tests are begun.

4.3.1.1 Temperature extremes. The power unit shall be operated in a temperature range of 160 to -65°F.

4.3.1.2 Flow rate. The flow rate through the power unit shall vary from 2.6 to 1 gal/min.

4.3.2 Mounting sample. For performance tests (3.4.1 through 3.4.4) mounting and orientation of the sample will be as in the vehicle. The longest dimension will be the longitudinal dimension.

4.3.3 Test apparatus accuracy. Accuracy of test apparatus shall be as specified in MIL-STD-810.

4.4 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of section 3 as listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. The sample units shall be examined for the defects specified in table II.

4.5 Conformance inspection (CI). CI shall consist of the examinations of 4.5.1 and the acceptance tests of 4.5.2.

4.5.1 Examination. The sample units shall be examined for the defects specified in table II.

4.5.2 Acceptance tests. The sample units shall be subjected to the acceptance tests specified in table I except fluid output requirement. All power units shall be tested for fluid output performance.

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TABLE I. Classification of inspections.

Title	Require-ment	Inspec-tion	First article		Conformance	
			PPI	IPI	Examin-ation	Accept-ance
Materials and construction	3.2, 3.2.1, 3.3.2, 3.3.6 and 3.3.8	4.6.1	X	X		
Defects (see 4.6.2 and table II)	3.3, 3.3.1, 3.3.3, 3.3.4, 3.3.5 thru 3.3.8, 3.5 thru 3.7	4.6.2	X	X	X	
Fluid output	3.4.1	4.6.3	X	X		X 1/
Relief valve	3.4.1.1	4.6.3.1	X	X		X
Leakage	3.4.1.2	4.6.3.2	X	X		X
Slope operation	3.4.2	4.6.4	X	X		X
Solenoid valve operation	3.4.3	4.6.5	X	X		X
EMI	3.4.4	4.6.6	X	X		X
Endurance	3.4.5	4.6.7	X	X		
High temperature	3.4.6.1	4.6.8.1	X	X		
Low temperature	3.4.6.2	4.6.8.2	X	X		
Humidity	3.4.6.3	4.6.8.3	X	X		
Steam and waterjet cleaning	3.4.6.4	4.6.8.4	X	X		
Shock	3.4.6.5	4.6.8.5	X	X		
Vibration	3.4.6.6	4.6.8.6	X	X		
Fungus resistance	3.4.6.7	4.6.8.7	X	X		
Corrosion resistance	3.4.6.8	4.6.8.8	X	X		
Sand and dust	3.4.6.9	4.6.8.9	X	X		

1/ 100% testing for fluid output performance.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Nonconformance in design and construction (see 3.3).	Visual
102	Dimensions affecting interchangeability out of tolerance (see 3.3).	SIE 1/
103	Faulty workmanship affecting performance (see 3.7).	Visual

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TABLE II. Classification of defects - Continued.

Category	Defect	Method of examination
Minor:		
201	Dimensions not affecting interchangeability out of tolerance (see 3.3).	SIE
202	Improper finish (see 3.5).	Visual
203	Improper marking (see 3.6).	Visual
204	Weight not as specified (see 3.3.7).	SIE
205	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2, 3.2.1, 3.3.2, 3.3.6 and 3.3.8, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2.1, 3.3, 3.3.1, 3.3.3 through 3.3.8 and 3.5 through 3.7, shall be determined by examination for the defects listed in table I. Examination shall be visual, or by measurement with standard inspection equipment.

4.6.3 Fluid output. To determine conformance to 3.4.1, when filled with fluid conforming to 3.3.3 and energized with 24 Vdc, the power unit shall provide no less than 1 gal/min at 1000 psig and no less than 2 gal/min at 500 psig.

4.6.3.1 Relief valve. To determine conformance to 3.4.1.1, it shall be verified that the relief valve in the pump discharge line shall begin to open at 2000 $\pm$100 psig.

4.6.3.2 Leakage. To determine conformance to 3.4.1.2, during the performance and environmental tests, it shall be verified that there are no leaks or drips. Weeps or seeps are allowable. With 500 psig pressure applied to the “p”, pressure port side, of the power unit when the solenoid is de-energized, fluid leakage shall not exceed 0.2 in^2 in 10 seconds before and after the endurance test.

4.6.4 Slope operation. To determine conformance to 3.4.2, it shall be verified that flow outputs and pressures specified in 3.4.1 are maintained when the unit is inclined longitudinally at $\pm$66% and laterally at $\pm$66% from the horizontal.

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4.6.5 Solenoid valve operation. To determine conformance to 3.4.3, it shall be verified that the solenoid valve, when energized with 18 to 30 Vdc, reverses the direction of flow between tank port and pressure port at both 500 and 2000 psig, and that the valve shifts back when voltage is removed.

4.6.6 Electromagnetic interference. To determine conformance to 3.4.4, the power units shall be tested in accordance with MIL-STD-462. If the EMI filter 12359539 is required, the power unit equipped with filter shall be installed in a vehicle and tested as specified in MIL-STD-461. Unless otherwise specified, vehicle testing shall be the responsibility of the procuring activity (see 6.2).

4.6.7 Endurance. To determine conformance to 3.4.5, the hydraulic power unit shall be operated intermittently for 1500 cycles from the fully extended to fully retracted positions with 5 minutes between cycles. During the test the unit shall operate the ramp or a simulated ramp of equivalent weight.

4.6.7.1 Endurance details. Intermittent operation is defined as operation for 5 minutes followed by non-operation for 5 minutes. A cycle is defined as operation of the unit from fully retracted position to fully extended position and return.

4.6.8 Environmental conditions. The environmental tests of 4.6.8.1 through 4.6.12 shall be conducted to determine conformance to 3.4.6. The performance tests of 4.6.3 through 4.6.6 shall be performed on all samples which have satisfactorily undergone the environmental tests. Verify that there is no degration of performance.

4.6.8.1 High temperature. To determine conformance to 3.4.6.1, the hydraulic power unit shall be tested according to MIL-STD-810D, method 501.3, procedures I and II within the following limits and guidelines. The storage temperature shall be 160°F, held constant for 48 hours. Following storage the power unit shall be operated intermittently for 1000 cycles in accordance with 4.6.3, at a stabilized 160°F.

4.6.8.2 Low temperature. To determine conformance to 3.4.6.2, the hydraulic power unit shall be tested per MIL-STD-810D, method 502.3, procedures I and II within the following limits and guidelines. The storage temperature shall be -70°F, held constant for 48 hours. Following storage the power unit shall be operated intermittently for 100 cycles in accordance with 4.6.3, at a stabilized -65°F.

4.6.8.3 Humidity. To determine conformance to 3.4.6.3, the hydraulic power unit shall be tested in accordance with MIL-STD-810D, method 507.3, procedure III. Near the end of the fifth and tenth cycles with humidity remaining at 95%, operate hydraulic power unit and record results (see figure 2).

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4.6.8.4 Steam and waterjet cleaning. To determine conformance to 3.4.6.4, the hydraulic power unit shall be steam cleaned, using high pressure steam. After steam cleaning, the unit shall be waterjet washed. The method for cleaning shall be as follows: The jet shall be applied perpendicular to the surface being cleaned, at a distance of not more than one foot (ft) for steam cleaning and not more five ft for waterjet cleaning. The jet pressure shall be 100 to 110 psig and the jet shall be moved at such a rate that the exposed surface of the power unit is subjected to the jet at a rate not to exceed one square foot per minute (ft^3/min). After testing, the power unit shall be tested as specified in 4.6.3 through 4.6.5.

4.6.8.5 Shock. To determine conformance to 3.4.6.5, the hydraulic power unit shall be tested in accordance with MIL-STD-810, method 516.4, procedure I, figure 516.4-4, 40 g peak value, 11 ms duration. The shock shall be applied to the face of the foot mount opposite from the unit. Following the test the unit shall have no deformed, cracked, or broken parts (see figure 3).

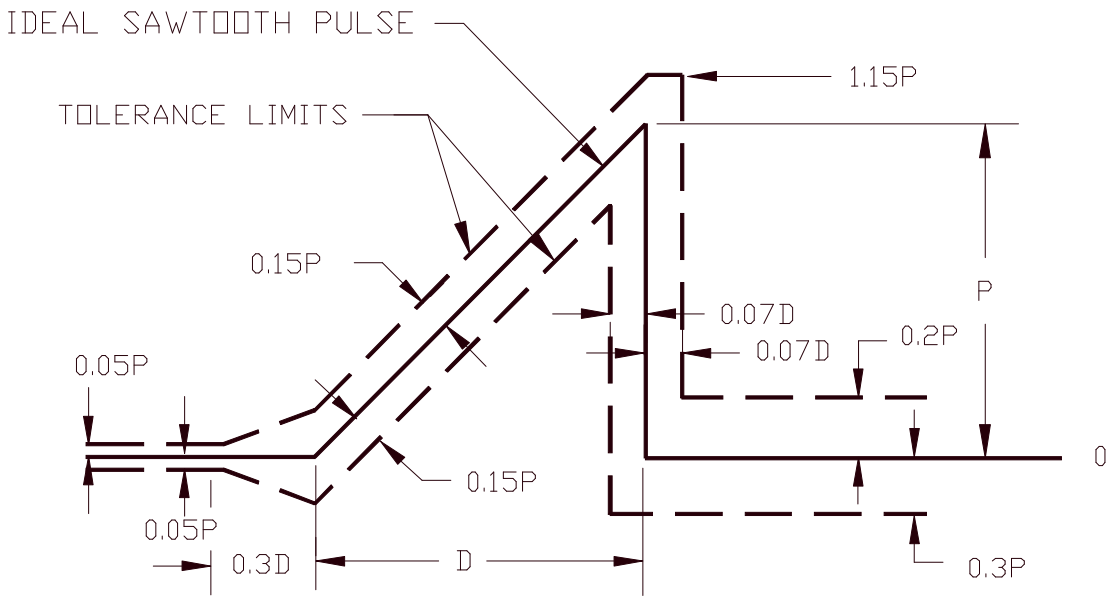
4.6.8.6 Vibration. To determine conformance to 3.4.6.6, the hydraulic power unit shall be tested in accordance with MIL-STD-810D, method 514.4, category 8, table 514.4-VII, and figure 514.4-35. Additional test conditions shall be as follows:

a. Sinusoidal vibration tests shall be conducted within the limits of figure 4.
b. Maximum load along the longitudinal axis shall be 10 g.
c. During the test, operation of the unit is unnecessary.
d. The unit shall be vibrated along each axis in accordance with the test levels, frequency range and times from figures 4 and 5.
e. Following the vibration tests the unit shall have no deformed, cracked or broken parts.

4.6.8.7 Fungus resistance. To determine conformance to 3.4.6.7, the hydraulic power unit shall be tested in accordance with MIL-F-13927, class 2, method B except that the test period shall be a continuous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without distributing any fungus growth that have developed. Upon completion of the examination and replacement of access port covers, the unit shall be operated intermittently for one hour, in accordance with 4.6.3.

4.6.8.8 Corrosion resistance. To determine conformance to 3.4.6.8, the hydraulic power unit shall be tested for resistance to corrosion in accordance with MIL-STD-202, test condition A of method 101D. Following the test, unit shall be operated intermittently for one hour in accordance with 4.6.3.

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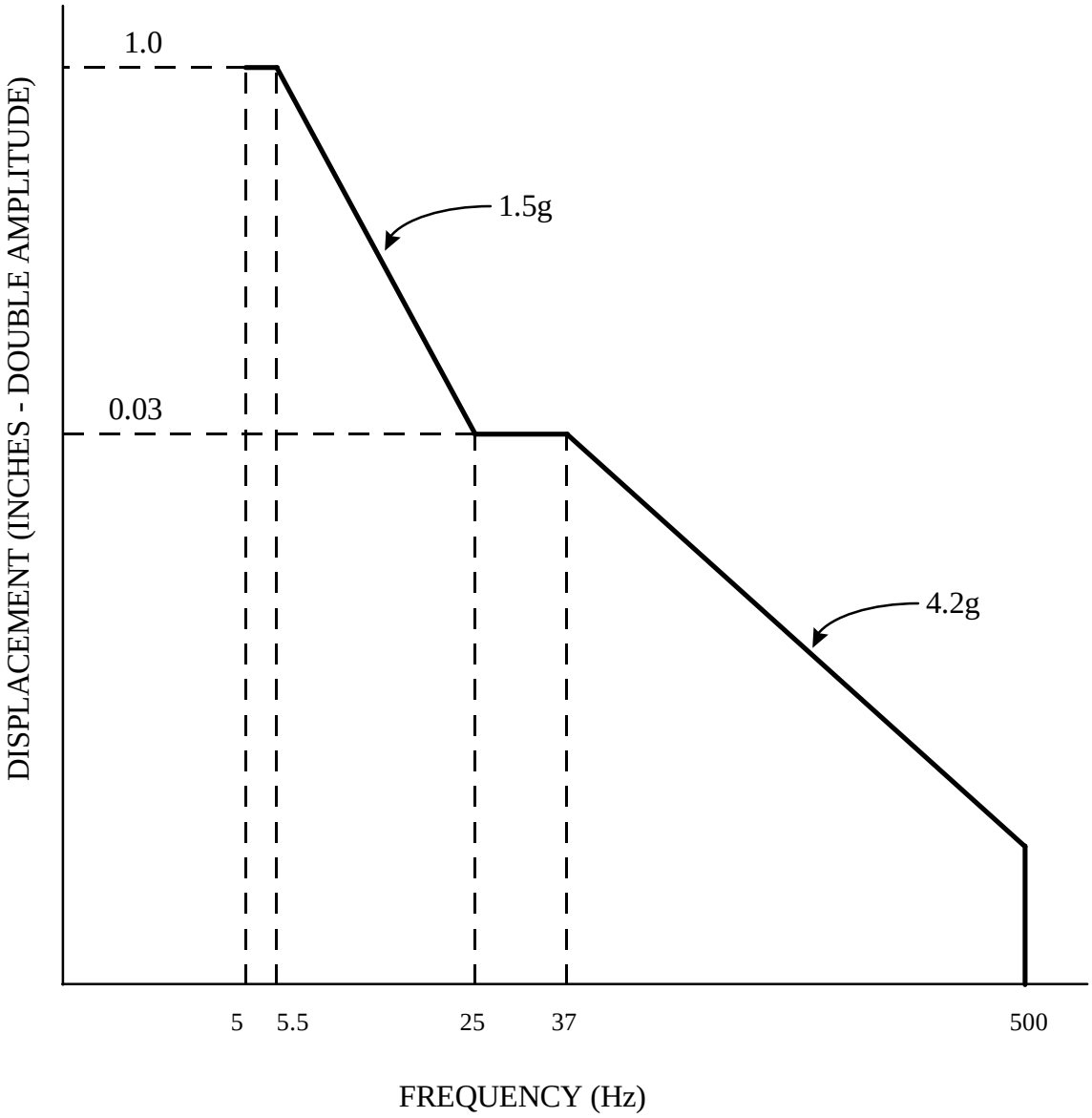


P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE: The oscillation shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 3. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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NOTES:

1. Acceleration levels: ±9 g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 4. Vibration test curve.

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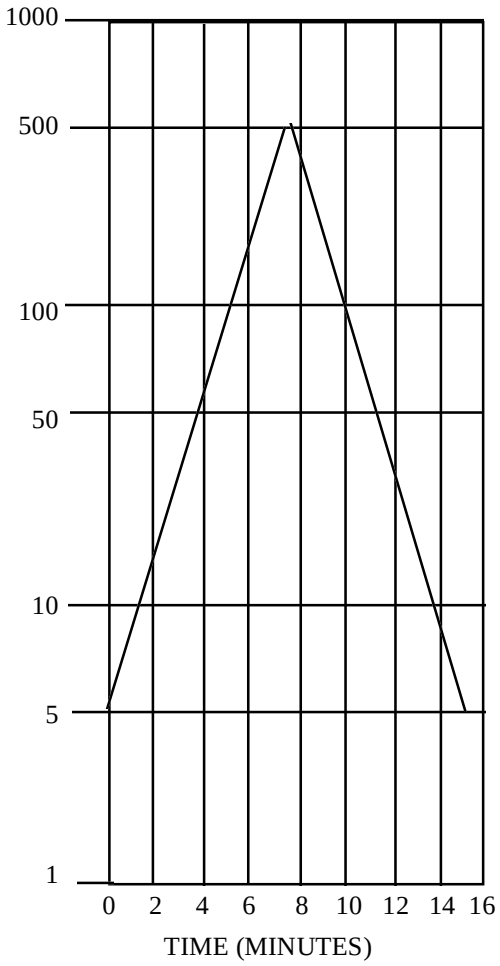


FIGURE 5. Logarithmic sweep.

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4.6.8.9 Sand and dust. To determine conformance to 3.4.6.9, the hydraulic power unit shall be subjected to the sand and dust test of MIL-STD-202, method 110A. During step 1 and 3 of the procedure the unit shall operate intermittently at full stroke and at a no load condition. Intermittent operation is described in 4.6.7.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The power unit manufactured in accordance with this specification is intended for ramp operation on vehicles of the Fighting Vehicle System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2).
c. If first article is required (see 3.1).
d. If PPI sample size is other than as specified (see 4.2.1).
e. If IPI sample size is other than as specified (see 4.2.2).
f. If inspection conditions are to be other than as specified (see 4.3).
g. If responsibility for vehicle testing to verify EMI requirement is to be other than as specified (see 4.6.6).
h. Packaging requirements (see 5.1).

6.3 Definition.

a. Cycle. A cycle is defined as operation of the unit from fully retracted position to fully extended position and return.

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b. Drip. Any evidence of fluid beyond a seal or splitline where droplets form and fall.
c. Intermittent operation. Intermittent operation is defined as operation for 30 seconds and non-operation for 2-1/2 minutes.
d. Leak. Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.
e. Seep. Any evidence of fluid beyond a seal or splitline that does not result in the formation of a droplet.
f. Weep. Any evidence of fluid beyond a seal or splitline.

6.4 Subject term (key word) listing.

Discharge ports
Flow rate
Fluid output
Relief valve
Solenoid valve

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 4320-0030)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62495B (AT)	**2. DOCUMENT DATE (YYMMDD)** 970221

3. DOCUMENT TITLE POWER UNIT, HYDRAULIC

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